EvidenceScene: Observability-Conditioned Pose Tracking and Revisable 3D Scene Reconstruction for Robot Manipulation

Tiwater
research@tiwater.com

***Abstract*—**Model-free 6D pose tracking and online surface reconstruction from RGB-D video remain brittle when depth support is sparse, objects self-occlude, or local geometry is degenerate. A common assumption is that returning more poses, or lowering an internal optimization residual, yields a more reliable reconstruction. We show that this assumption fails. Across a 31-framework audit, repairing two Point2Pose implementation defects on Laval Cookiejar raises pose availability from 207 to 239/295 frames and failure-inclusive ADD AUC from 0.574 to 0.635, yet reduces 20 mm origin successes from 136 to 120 and drops 10 mm surface coverage C_{10} from 76.82% to 61.79%. We trace this availability–surface paradox to three coupled mechanisms: unanchored SE(3) gauge drift in disconnected graph components, nuisance-coupled rank deficiency after landmark elimination, and pose-version desynchronization between graph revisions and fused surfaces. We present EvidenceScene, a standalone, causal (no future-frame) framework that commits only sensor-supported, observable, version-consistent state from registered RGB-D and SAM 2 masks. Auditing Point2Pose under four modular interventions isolates prior-connected admission (G), marginal rank gating (R), immutable feature-birth ledgers (M), and surface-history transactions (S); on our native BundleTrack pose owner, deterministic CUDA RANSAC and exact SE(3) inverse transfer raise Clock/Cookiejar strict successes from 135/107 to 191/136 (30.24 ms p_{50}, 33.1 fps on YCBInEOAT), while guarded `NO_BA` retirement saves 928.4–1,994.0 MiB RSS with byte-identical outputs. A columnar lineage-indexed `RevisableSurfelMap` and two-stage `support10/coarse12mm` selector raise 1,500-point C_{10} from 40.57%/52.34% to 85.10%/85.05% while accelerating 60k-point update (73.00 → 1.05 ms), snapshot (20.48 → 0.22 ms), and selection (15.52 → 1.05 ms; 8.65× geomean speed-up). Finally, bit-exact runtime adapters overlap SAM 2 GPU waits with CPU preparation and use O(runs) RLE arithmetic, cutting warm endpoint p_{50}/p_{95} from 155.33/219.06 ms to 127.80/153.40 ms at 499/499 parity.

***Index Terms*—**EvidenceScene, evidence-constrained 3D scene reconstruction, RGB-D object tracking, revisable surface reconstruction, factor-graph observability, degenerate geometry, robot manipulation, bit-exact runtime optimization, reproducibility.

I. Introduction

ONLINE 6-DoF object pose tracking and 3D surface reconstruction from onboard RGB-D streams provide essential spatial awareness for remote humanoid teleoperation and bimanual manipulation [1]–[3]. Without pre-loaded CAD models, a manipulation perception system must initialize an object coordinate gauge from the first segmented RGB-D observation, track relative rigid motion across occlusions and depth dropouts, and accumulate a geometrically trustworthy 3D surface for operator guidance.

A common assumption in model-free tracking benchmarks is that returning a higher fraction of non-lost poses—or achieving a lower internal optimization residual—naturally yields a more reliable 3D reconstruction. Our empirical investigation refutes this assumption. Under sparse depth support, self-occlusion, or locally degenerate geometry, long-horizon 2D point trajectories and dense correspondences frequently constrain only a subspace of the joint pose-landmark state. Admitting unconstrained or unanchored updates into persistent object maps and volumetric/surfel surfaces can corrupt previously verified geometry even while increasing frame-level pose availability.

To prevent implementation artifacts from being conflated with algorithmic gains, we introduce **EvidenceScene** (*Observability-Conditioned 3D Scene Reconstruction for Robot Manipulation*) and strictly distinguish three methodological entities throughout this study: (1) the upstream BundleTrack algorithm [1] (pinned to `ffa67d425240`), (2) the upstream Point2Pose point-trajectory tracker and TSDF reconstructor [2] (pinned to `51856226610d`), and (3) our proposed **EvidenceScene** framework, which combines observability-conditioned graph and surface transactions (F, G, R, M, S) on the Point2Pose branch with deterministic CUDA RANSAC, exact SE(3) inverse-pose transfer, guarded `NO_BA` frame retirement, lineage-audited `CausalObjectSurface` (`RevisableSurfelMap`) fusion, and bit-exact runtime execution on the BundleTrack branch. Candidate estimators execute in a shadow protocol, meaning that they are scored on identical registered RGB-D frames and SAM 2 [22] masks bound by SHA-256 receipts but are not published to the live teleoperation output. Throughout, *causal* means that an estimate at time t is a function of observations up to t; no future frame is consulted.

Evidence scope. All quantitative comparisons are development results on recordings available during method design. Pose scores use reference masks and reference-assisted initialization; surface scores use a CAD or sampled-model reference. They are not a held-out or physical-ground-truth evaluation, and the Point2Pose branch is an offline shadow; no result authorizes a live-estimator cutover.

Our contributions are fourfold:

1) **Failure Mechanism Formulation and Observability Audit (Sections III–IV):** We formalize three coupled

Tiwater (e-mail: research@tiwater.com). Project page: https://tiwater.github.io/evidencescene/.

failure modes in long-horizon RGB-D tracking and fusion: (i) temporal attribution and solver-return inconsistency (Fork F), (ii) unanchored $\mathrm{SE}(3)$ gauge drift and nuisance-coupled rank deficiency after orthogonal left-null-space landmark elimination $\bar{\mathbf{A}}_x = (\mathbf{I} - \mathbf{A}_y \mathbf{A}_y^{+})\mathbf{A}_x$, and (iii) pose-version desynchronization between factor-graph revisions and accumulated surfaces. We evaluate controlled interventions (G, R, M, S) across 704 Vicon-referenced frames and 71 exact native factor-graph replays, and report regressions alongside gains.

2) **Native BundleTrack Determinism, Precision, and Memory Governance (Section IV-B):** We identify and fix parallel-reduction tie-breaking races and count-comparison directions in native CUDA RANSAC, repair float-to-matrix inverse-coefficient truncation in `CUDACache::storeFrame`, and introduce guarded `NO_BA` non-keyframe retirement that reclaims 928.4–1,994.0 MiB of host RSS under tracking loss while preserving 100% bit-exact pose and surface outputs.
3) **Lineage-Audited Surface Fusion and Columnar Budget Selection (Section IV-C):** By tracing 1,512,357 surfel contributions to 1,960,512 source pixels and continuous mesh rays, we separate pose-induced surface tails from raw boundary measurements, and implement a columnar `RevisableSurfelMap` and array-native `support10/coarse12mm` 1,500-point selector that raises model coverage C_{10} from $40.57\%/52.34\%$ to $85.10\%/85.05\%$ while achieving an $8.65\times$ geometric-mean kernel speed-up ($93.0\times$ snapshot, $69.8\times$ update, $14.8\times$ selection) with bit-identical digests.
4) **31-Framework Context and Bit-Exact Runtime Execution Optimization (Sections IV-D–V, Appendix A):** In a 31-framework audit whose full per-framework benchmark table appears in Appendix A (Table IX), we evaluate depth foundation models, pointmap transformers, segmenters, trackers, and matchers on identical registered inputs; EfficientLoFTR, XFeat, and ALIKED+LightGlue are retained as shadow trials. We further introduce `SAM2GpuWindowOverlapScope` and `RequestEvidenceDeduplicationScope` ($O(\text{runs})$ exact integer RLE arithmetic, cross-stage surface/medoid reuse, BLAS-screened coplanar RANSAC, and request-boundary cyclic GC control), cutting warm HTTP endpoint latency by 27.6% ($176.40 \rightarrow 127.80$ ms p_{50}, $219.06 \rightarrow 153.40$ ms p_{95}) with 100% bit-exact parity.

II. Related Work and Architectural Landscape

Model-Free 6-DoF Pose Tracking, Point Tracking, and Feature Matching: Model-free RGB-D trackers estimate camera-local object motion from an initial segmented frame for manipulation tasks [44], [45]. BundleTrack [1] combines pairwise feature matching with keyframe pose-graph optimization over sparse 3D correspondences and dense point-to-plane residuals. BundleSDF [3] jointly optimizes a keyframe pose graph and a neural implicit signed distance field, whereas FoundationPose [4] and Any6D [40] unify CAD-based and few-view reference-assisted pose estimation, alongside iterative, Gaussian Splatting [27], and KV-cache trackers such as BIT [41], 6DOPE-GS [42], and KV-Tracker [43]. Point2Pose [2] introduces long-horizon 2D point tracking (Online BootsTAPIR [9], alongside TAPNext++ [10] and CoTracker3 [11]) across occlusions, lifting 2D tracks to 3D observations for multi-hypothesis SVD-RANSAC [24], [25] registration, GTSAM [7] / iSAM2 [8] Levenberg-Marquardt pose-landmark graph optimization, and incremental TSDF fusion. While Point2Pose's point identity persistence, multi-object query management, and graph/TSDF architecture are established upstream contributions, we show that its unguarded graph commits and decoupled TSDF integration can degrade surface fidelity when 2D tracks lack well-distributed 3D depth support. In parallel, replacing detector-free LoFTR [12] or dense RoMa [17] with faster matchers such as EfficientLoFTR [13], semi-dense XFeat [14], or sparse ALIKED+LightGlue [15], [16] exposes a strict accuracy–latency frontier when foreground masks contain $< 1\%$ valid sensor depth.

Depth Foundation Models, 3D Pointmaps, and Open-World Segmentation: Monocular relative and metric depth estimators—including Depth Anything V2 [28], Prior Depth Anything (Prior-DA) [29], Apple Depth Pro [30], and UniDepthV2 [31]—alongside feed-forward multi-view pointmap transformers—VGGT [32], SLAM3R [33], DUSt3R [34], MASt3R [35], and CUT3R [36]—predict dense geometry from RGB images, while single-image 3D object generators such as SAM 3D Objects [23] synthesize complete 360° object shape priors. For open-vocabulary grounding and temporal mask tracking, Grounding DINO [37], Florence-2 [38], SAM 2 [22], and distilled YOLO11 [39] segmenters provide instance masks that must be isolated across object roles to prevent memory contamination.

Observability under Degeneracy, Revisable Surface Mapping, and Kinematic Calibration: State estimation under geometric degeneracy has been studied in point-cloud registration and SLAM by Zhang *et al.* [18] and Tuna *et al.* (X-ICP [19]), which project optimization increments onto well-conditioned Hessian subspaces. However, evaluating pose conditioning with landmarks held fixed overestimates available information when the object map itself is uncertain. In dense RGB-D mapping—spanning point-based surfel fusion [21], Open3D [26], and BundleFusion [20]—BundleFusion de-integrates and re-integrates RGB-D frames when bundle adjustment revises camera poses; our `RevisableSurfelMap` adds an explicit birth-keyframe index so that a pose revision becomes a surface transaction rather than an in-place overwrite. For unknown dynamic objects, however, re-integrating historical keyframes with revised graph poses only improves the surface if the graph revisions themselves are metrically accurate and anchored to the initial gauge. Finally, for bimanual pouring [46] and multi-camera-to-61-link calibration, we extend $\mathbf{AX} = \mathbf{XB}$ hand-eye calibration [47] and AprilTag fiducial PnP [48] on Lie groups [49] into a MoCap-free bilateral kinematic closure protocol.

III. Problem Formulation and Failure Mechanisms under Degenerate Observations

A. Observation Provenance, Coordinate Gauge, and Metrics

Let each tracking observation tuple be bound as

$$\mathcal{O}_t = \big(\text{obj_id}, f_t, \{(u_j, v_j, d_j, \text{id}_j, \text{valid}_j)\}_{j=1}^{M_t}, \mathbf{K}, M_t^{\text{seg}}\big), \tag{1}$$

distinguishing sensor-measured 3D points $\mathbf{z}_{t,j} = \pi^{-1}(u_j, v_j, d_j; \mathbf{K})$ from depth-missing 2D tracks, RANSAC proposals, numerical initializations, and the fixed Frame-0 gauge O_{obj_0}. Each tracker's initial gauge is aligned once using the first reference pose, with Frame 0 excluded from score denominators. For post-initial frames, ADD measures mean corresponding model-vertex distance and ADD-S measures mean nearest-vertex distance; unavailable outputs receive zero recall at all thresholds in 0–100 mm normalized recall AUC.

For an accumulated reconstructed point set S and reference surface point set M evaluated on a common $4\,\text{mm}$ grid without ICP drift masking, we simultaneously report directional precision P_ϵ and reference coverage C_ϵ:

$$P_\epsilon = |\{s \in S : \min_{m \in M} \|s - m\|_2 \le \epsilon\}|/|S|, \tag{2}$$

$$C_\epsilon = |\{m \in M : \min_{s \in S} \|m - s\|_2 \le \epsilon\}|/|M|, \tag{3}$$

alongside directional p_{95} distances F_{95} ($S \to M$) and R_{95} ($M \to S$). Deleting uncertain points can inflate P_ϵ while collapsing C_ϵ; a valid surface improvement must account for both directions.

B. Mechanism 1: Temporal Attribution and Solver-Return Inconsistency

Source inspection of pinned Point2Pose (`51856226610d`) reveals two internal state-contract defects:

1) *Historical Keyframe Attribution Overwrite:* When a pending keyframe from historical frame $f_{\text{kf}} < f_t$ is promoted after multi-frame verification, `ModularPipeline.step` writes the graph-optimized keyframe pose $\mathbf{T}_{\text{kf}}$ into current `obj.pose` without checking $f_{\text{kf}} == f_t$.
2) *Refined-Pose Return Mismatch:* In `SVDClusterRANSACRegister.register`, when local refinement updates a hypothesis, the function recomputes final inliers and residuals using refined $\mathbf{T}_{\text{ref}}$ but returns pre-refinement $\mathbf{T}_{\text{raw}}$.

C. Mechanism 2: Unanchored Gauge Drift and Nuisance-Coupled Rank Deficiency

In GTSAM [7], numerical `Values` provide linearization points, whereas only `Factors` constrain the state. Native Point2Pose places a single prior factor on initial keyframe pose $\mathbf{X}_0 \in \text{SE}(3)$ and connects subsequent poses $\mathbf{X}_m$ ($m \in \mathcal{V}_p$) and 3D landmarks $\mathbf{p}_n \in \mathbb{R}^3$ ($n \in \mathcal{V}_\ell$) via `BearingRangeFactor`s (from registration inliers) and conditional `BetweenFactor`s (admitted only when all-correspondence mean residual $< 0.01\,\text{m}$).

Unanchored Gauge Freedom: Let $C_0 \subseteq \mathcal{V}_p \cup \mathcal{V}_\ell$ denote the factor-graph connected component containing the sole initial prior $\mathbf{X}_0$. For any disconnected component $C' \cap C_0 = \emptyset$, transforming all inverse poses and landmarks in C' by an arbitrary global rigid transformation $\mathbf{Q} \in \text{SE}(3)$, $\mathbf{X}'_m = \mathbf{Q}\mathbf{X}_m$ and $\mathbf{p}'_n = \mathbf{Q}\mathbf{p}_n$, leaves every relative observation $(\mathbf{X}'_m)^{-1}\mathbf{p}'_n = \mathbf{X}_m^{-1}\mathbf{p}_n$ and relative-pose factor invariant. Thus Levenberg-Marquardt damping can converge numerically with non-increasing cost while C' drifts freely relative to O_{obj_0} and overwrites persistent frontend map coordinates.

Nuisance-Eliminated Local Observability: Even when $\mathbf{X}_i \in C_0$, topological connectivity does not imply 6-DoF observability. In the whitened, prior-conditioned local linear system around the current linearization point, let $\delta\mathbf{x} \in \mathbb{R}^{6|\mathcal{V}_p|}$ denote pose tangent increments and $\delta\mathbf{y} \in \mathbb{R}^{3|\mathcal{V}_\ell|}$ denote landmark increments:

$$\min_{\delta\mathbf{x}, \delta\mathbf{y}} \|\mathbf{A}_x \delta\mathbf{x} + \mathbf{A}_y \delta\mathbf{y} - \mathbf{b}\|_2^2. \tag{4}$$

Eliminating uncertain landmarks via orthogonal projection onto the left null space of $\mathbf{A}_y$ yields the reduced pose Jacobian $\bar{\mathbf{A}}_x = (\mathbf{I} - \mathbf{A}_y\mathbf{A}_y^+)\mathbf{A}_x$ and marginal pose information matrix:

$$\bar{\mathbf{H}}_x = \bar{\mathbf{A}}_x^\top \bar{\mathbf{A}}_x = \mathbf{A}_x^\top(\mathbf{I} - \mathbf{A}_y\mathbf{A}_y^+)\mathbf{A}_x \preceq \mathbf{A}_x^\top \mathbf{A}_x. \tag{5}$$

Treating uncertain landmarks as known ($\mathbf{A}_x^\top \mathbf{A}_x$) strictly overestimates pose conditioning. The locally constrained dimension of current pose i is given by the column-deletion rank difference:

$$r_i = \text{rank}(\bar{\mathbf{A}}_x) - \text{rank}(\bar{\mathbf{A}}_{x,-i}) \in \{0, \dots, 6\}, \tag{6}$$

where $\bar{\mathbf{A}}_{x,-i}$ removes the six columns of pose i. Under non-collinear 3D correspondences anchored to C_0, one shared landmark supplies $r_i = 3$, two supply $r_i = 5$, and three non-collinear landmarks supply $r_i = 6$.

D. Mechanism 3: Pose-Version Desynchronization in Surface Fusion

When `KeyFrameGraph.update` revises a historical keyframe pose from its initial integration estimate $\mathbf{T}_\tau^{(\tau)}$ to $\mathbf{T}_\tau^{(t)}$ ($t > \tau$), `ModularPipeline.step` updates the sparse landmark map, but `SDFBuilder.integrate_keyframe` only integrates the newly promoted keyframe into the TSDF volume without de-integrating or re-projecting earlier observations fused at stale poses $\{\mathbf{T}_\tau^{(\tau)}\}$. Consequently, the sparse tracker map and dense TSDF surface diverge in coordinate version history.

IV. The EvidenceScene Framework: Constrained State Updates, Native Governance, and Zero-Loss Execution

Fig. 1 summarizes the framework. The native BundleTrack branch owns the live pose and surface via deterministic RANSAC winner selection, exact SE(3) inverse transfer, guarded `NO_BA` retirement, and a lineage-indexed surfel map. The Point2Pose branch is an offline shadow that isolates graph and surface failure mechanisms through F, G, R, M, S; its outputs are scored but never published. We call an intervention

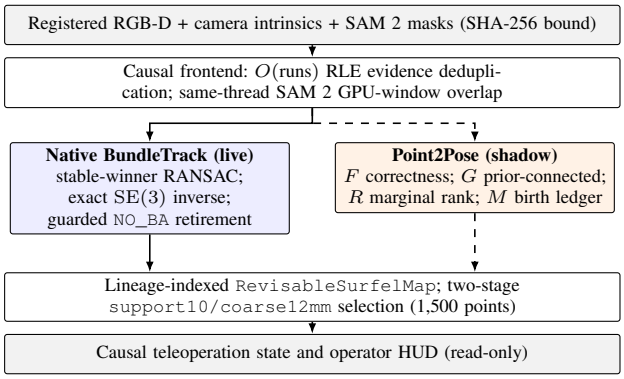


Fig. 1. **Framework overview.** Both branches consume the same registered RGB-D stream and SAM 2 masks. The native BundleTrack branch owns the live pose and surface; the Point2Pose branch is an offline shadow used to isolate graph and surface failure mechanisms through F, G, R, M, S. Solid arrows: live authority; dashed: shadow branch (scored, never published). The selector emits at most 1,500 points to the operator interface.

observability-conditioned when its commit decision depends on a measured conditioning test of the local normal equations, not only on optimizer cost.

A. Controlled Graph and Surface Interventions (F, G, R, M, S)

Within **EvidenceScene**, we first isolate five modular, default-off interventions on the Point2Pose causal trajectory architecture:

- **Correctness Baseline (F):** Enforces frame-scoped pose attribution ($f_{\text{kf}} == f_t$ before updating `obj.pose`) and returns refined pose $\mathbf{T}_{\text{ref}}$ consistently with its inliers and residuals.
- **Prior-Connected Graph Admission (G):** Admits optimized keyframe and landmark updates into the frontend map only for variables residing in the connected component C_0 of initial pose prior $\mathbf{X}_0$. Disconnected components retain their optimizer state for potential later reconnection without corrupting the active map.
- **Marginal Rank Gating (R):** Extends G by computing r_i in (6) across three numerical tolerances $(\{0.1\times, 1\times, 10\times\}\epsilon \max(\text{shape}(\bar{\mathbf{A}}_x))\sigma_{\max}(\bar{\mathbf{A}}_x))$ and withholding graph commit when current pose $r_i < 6$.
- **Feature-Birth Observation Ledger (M):** Records immutable original camera-space 3D measurements $(\text{obj_id}, f_{\text{birth}}, f_{\text{admit}}, \text{track_id}, \mathbf{z}_{\text{cam}})$ when features are first sampled, and inserts deduplicated native `BearingRangeFactors` at the birth pose once registration instantiates the landmark.
- **Causal Surface-History Transaction (S):** Maintains an immutable ledger of integrated depth/RGB observations and their pose versions. Whenever an admitted graph update modifies a previously integrated keyframe pose, S deterministically rebuilds a temporary volume from causal history using current pose versions $\mathbf{T}_\tau^{(t)}$ before integrating new evidence, committing atomically upon grid and identity verification.

B. BundleTrack Native Determinism, Inverse Precision, and Frame Retirement

Auditing our primary BundleTrack C++/CUDA pose estimator and session adapter uncovered four concrete engineering improvements:

1) **`NO_BA` Status Integrity:** A two-line adapter guard prevents retained native `NO_BA` states from being misclassified as historical candidates on cache misses, eliminating 6 false historical promotions while preserving all $34/38$ valid task candidates.
2) **CUDA RANSAC Deterministic Winner Selection:** In `BundleTrack/src/cuda/`, two identical 13-correspondence CPU PCL inputs produced divergent 6-inlier subsets differing by $18.00\,\text{mm}$ and $6.22°$ due to parallel atomic tie-breaking and reversed count-comparison inequalities. Applying the `stable-winner` and count-direction repair collapses 100 repeated native RANSAC+Procrustes calls onto a single deterministic output.
3) SE(3) **Inverse-Coefficient Precision in `storeFrame`:** Replacing lossy bulk cast transfer with exact element-wise Eigen-to-`float4x4` inverse-pose coefficient copying in `CUDACache::storeFrame` ensures pinhole backprojection accuracy $\leq 1.30 \times 10^{-7}\,\text{m}$.
4) **Guarded `NO_BA` Non-Keyframe Retirement:** Under continuous tracking loss, `Bundler` previously retained all blocked frames in GPU/host memory. We invoke `Bundler::forgetFrame` on non-keyframes beyond the window_size $+ 1$ budget only when the most recent window_size $+ 1$ frame IDs are contiguous and all carry status `NO_BA`, while protecting first, current, neighbor, local, keyframe, and pose-cached frames.

C. Preparation-Stage Generative 3D Prior Anchoring, Columnar Surfel Fusion, and Surface Selection

During manipulation, cameras observe only the visible front hemisphere of grasped and target objects, leaving self-occluded back surfaces unobserved until sufficient viewpoint rotation occurs. When a single-image 3D generative shape prior (such as Meta's SAM 3D Objects [23]) is available during the non-real-time *Stage-0 preparation phase* ($t = 0$), **EvidenceScene** constructs a complete 360° object prior $\mathcal{P}_{\text{prior}} = \{\mathbf{q}_j\}_{j=1}^{N} \subset \mathbb{R}^3$ from the initial SAM 2 instance mask and anchors its monocular scale $s > 0$, rotation $\mathbf{R} \in \text{SO}(3)$, and translation $\mathbf{t} \in \mathbb{R}^3$ to the calibrated frame-0 RGB-D front-surface point cloud $\mathcal{P}_{\text{meas}}^{(0)} = \{\mathbf{p}_i\}_{i=1}^{M}$ via trimmed Umeyama Sim(3) registration [24]:

$$(s^\star, \mathbf{R}^\star, \mathbf{t}^\star) = \arg \min_{s>0, \mathbf{R}\in\text{SO}(3), \mathbf{t}} \sum_{i\in\mathcal{I}_\tau} \left\| s\mathbf{R}\mathbf{q}_{\pi(i)} + \mathbf{t} - \mathbf{p}_i \right\|_2^2, \tag{7}$$

where $\mathcal{I}_\tau$ retains the $\tau = 85\%$ closest front-surface correspondences so unobserved back-side vertices on $\mathcal{P}_{\text{prior}}$ do not bias the metric front-surface fit. During the real-time 30 Hz online loop, the anchored prior $\hat{\mathcal{P}}_{\text{prior}} = s^\star\mathbf{R}^\star\mathcal{P}_{\text{prior}} + \mathbf{t}^\star$ moves rigidly with the observability-gated keyframe pose $\mathbf{T}_{W\leftarrow O}(t)$, and `fuse_prior_with_measured_surface` enforces an *evidence-first replacement rule*: within every $r_c = 12\,\text{mm}$

coarse ball around a corroborated sensor surfel (support ≥ 2), measured surfels supersede prior points, whereas outside observed balls ($\min_{\mathbf{x}\in\mathcal{S}_{\text{meas}}} \|\hat{\mathbf{q}} - \mathbf{x}\|_2 > r_c$), prior points are retained with explicit provenance (support $= 0$, `is_prior = True`) to provide complete 360° scene geometry without fabricating sensor authority.

To transmit compact object surfaces within the 1,500-point output budget without sacrificing geometric coverage, we architect `RevisableSurfelMap` around a sorted columnar Structure-of-Arrays (SoA) engine with 1D bijective `int64` key-packing and an array-native `support10/coarse12mm` selector. Operating directly on contiguous lexicographically ordered coordinate-sum, support, and camera-count arrays, it prioritizes voxels with support ≥ 10 across 12 mm coarse cells before backfilling remaining capacity, cutting 60k-point update ($73.00 \rightarrow 1.05$ ms, $69.8\times$), snapshot ($20.48 \rightarrow 0.22$ ms, $93.0\times$), and selection ($15.52 \rightarrow 1.05$ ms, $14.8\times$) for an $8.65\times$ geometric-mean kernel speed-up with bit-identical digests.

D. Zero-Loss Runtime Execution: GPU-Window Overlap and $O(runs)$ Evidence Deduplication

Critical-path profiling of the synchronous `/world_model/observe` endpoint revealed that sequential SAM 2 and native BundleTrack execution ($95.87/118.98$ ms median/p_{95}) was compounded by ~ 48 ms of redundant host CPU work and mid-request cyclic GC stalls. We introduce two stacked, bit-exact perception adapters:

1) Same-Thread SAM 2 GPU-Window CPU Preparation Overlap (`SAM2GpuWindowOverlapScope`): On warm distorted frames, `SAM2VideoMemoryTracker.track` enqueues CUDA kernels for `propagate_in_video` in ~ 9.0 ms and blocks the host thread for ~ 24.2 ms at `(logits>0.0).cpu().numpy()` waiting for GPU `memory_attention`. Instead of spawning background `ThreadPoolExecutor` workers that contend for the Python GIL, our adapter performs a 0.25 ms fail-closed pre-validation, launches `propagate_in_video`, and executes mask-independent CPU preparation *on the same thread inside the* 24.2 ms *GPU wait window* before calling `.cpu().numpy()`: (i) bilinear/nearest undistortion `rectify_registered_rgbd`, (ii) dual SHA-256 evidence-hash prefixes covering `header+color_rgb` and `header+color_rgb+depth_mm` ($4.61/5.53$ MB $= 83.3\%$ of input bytes), and (iii) private `color_bgr` and `depth_m` buffers for `PinnedBundleTrackSession`. After GPU mask readback, it reuses in-scope `sensor_mask` RLE/SHA-256 and finalizes the remaining hash tail synchronously, eliminating all 38 per-frame background hash threads.

2) Request-Scoped Evidence Deduplication and O(runs) Run-Interval Arithmetic (`RequestEvidenceDeduplicationScope`):

- *Exact $O(runs)$ RLE Bounding Box and Centroid:* `FoundationMaskCacheDetector` previously iterated pixel-by-pixel in Python over 80,696 foreground pixels per mask, allocating over 161,000 Python `int` objects per frame (10.38 ms). Given RLE runs (s_k, L_k) with $e_k = s_k + L_k - 1$ on a $H \times W$ grid, let $(y_0, x_0) = \text{divmod}(s_k, W)$ and $(y_1, x_1) = \text{divmod}(e_k, W)$. Using exact Python integers, the row sum $\sum y$ across a multi-row run ($y_0 < y_1$) is evaluated in $O(1)$ closed form:
$$\Sigma_{y,k} = (W - x_0)y_0 + W\tfrac{(y_1-y_0-1)(y_0+y_1)}{2} + (x_1+1)y_1, \tag{8}$$
and the column sum follows from linear index sums $\Sigma_{\text{idx},k} = L_k s_k + L_k(L_k-1)/2$ via $\sum x = \sum_k \Sigma_{\text{idx},k} - W\sum_k \Sigma_{y,k}$, reducing box/centroid evaluation from 10.38 ms to 0.015 ms (99.85% reduction) with bit-for-bit identity.
- *Cross-Stage Surface and Medoid Reuse:* Within a request scope, we memoize `observation_digest`, `decode_compressed_depth`, and `decode_mask_rle`, and reuse Z-buffered `RegisteredRgbdFrame.select_surface(mask)` and `select_mask_centroid_depth_component` across `table_contexts`, `detector._position_detections_for_observation`, and `run_live._live_surface_projection_from_detection` ($41/41$ hits, 0 fallbacks).
- *Vectorized Coplanar AprilTag Plane RANSAC:* `fast_fit_shared_plane` caches deterministic `default_rng(0)` 240×3 index triplets by point count, screens candidate normals via BLAS matrix multiplication, and verifies top contenders in exact `float64`, reducing plane fitting on tag-visible frames from 4.47 ms to 0.89 ms.
- *Request-Boundary Cyclic GC Management:* Calling `gc.collect()` after model initialization reclaims $\sim$ 745 MB of unreferenced tensors (reducing CUDA allocated memory from 813 MB to 67.1 MB) and defers cyclic GC to request boundaries, eliminating 136–141 ms mid-request Generation-2 scans.

V. Experimental Evaluation and Mechanism Ablations

A. Metric-Dependent Pose Baselines and the Availability–Surface Paradox

Table I compares Point2Pose and BundleTrack on six reference-mask YCBInEOAT [5] sequences with identical 3×3 eroded masks. BundleTrack passes more frames at 10 mm ADD-S on all six sequences, whereas Point2Pose achieves higher 0–100 mm recall AUC on Cracker and Bleach-hard. On Sugar, the 10 mm success counts reverse between metrics: 617 vs. 794 under nearest-vertex ADD-S and 206 vs. 174 under corresponding-vertex ADD (Point2Pose vs. BundleTrack), showing that single-threshold rankings depend on symmetry and metric conventions.

On the three Vicon-referenced Laval actions [6] sharing identical SAM 2 masks and RGB-D entries ($203/205/296$ frames), factorial decomposition on Clock-hard confirms that frame attribution alone yields $43/202$ available poses (AUC 0.1923), solver-return consistency alone yields $158/202$ (AUC 0.6260), and combined fork F yields $157/202$ (AUC 0.6583) vs. original P at $43/202$ (AUC 0.1881). Yet Table II and Fig. 2 expose the central paradox on Cookiejar: while F

TABLE I
SELECTED REFERENCE-MASK YCBINEOAT DEVELOPMENT RESULTS. COUNTS ARE ADD-S SUCCESSES WITHIN 10 MM; AUC IS FAILURE-INCLUSIVE OVER 0–100 MM. P2P: POINT2POSE; BT: BUNDLETRACK. *BLEACH-HARD WAS SELECTED AFTER THE FIVE-SEQUENCE PROTOCOL.

Sequence	n	P2P 10	BT 10	P2P AUC	BT AUC
Cracker	374	365	373	0.961	0.952
Mustard	736	726	736	0.946	0.951
Bleach	662	155	301	0.798	0.851
Sugar	1001	617	794	0.915	0.920
Tomato	1307	1301	1303	0.957	0.959
Bleach-hard*	440	416	418	0.948	0.930

TABLE II
OPENED VICON-REFERENCE DEVELOPMENT SEQUENCES. COUNTS USE ALL POST-INITIALIZATION FRAMES; AUC COUNTS UNAVAILABLE POSES AS ZERO. B: BUNDLETRACK; P: ORIGINAL POINT2POSE; F: CUMULATIVE CORRECTNESS FORK. THESE ARE NOT HELD-OUT RESULTS.

Sequence	Method	Avail.	$\leq$10 mm	$\leq$20 mm	$\leq 10^\circ$	AUC
Clock-hard	B	202/202	119/202	199/202	162/202	0.887
Clock-hard	P	43/202	25/202	40/202	28/202	0.188
Clock-hard	F	157/202	27/202	137/202	122/202	0.658
Clock-full	B	197/204	150/204	197/204	197/204	0.885
Clock-full	P	112/204	64/204	101/204	60/204	0.458
Clock-full	F	169/204	156/204	169/204	85/204	0.721
Cookiejar	B	295/295	109/295	214/295	112/295	0.723
Cookiejar	P	207/295	54/295	136/295	140/295	0.574
Cookiejar	F	239/295	64/295	120/295	142/295	0.635

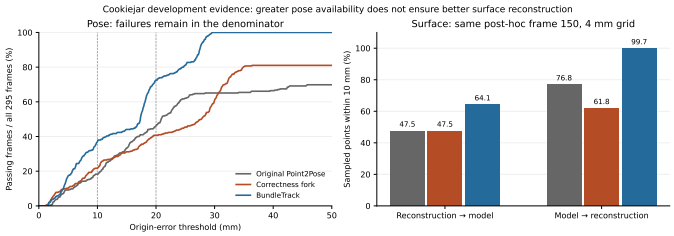


Fig. 2. **The availability–surface paradox on Laval Cookiejar.** Repair fork F increases pose availability ($207 \rightarrow 239/295$) and ADD AUC ($0.574 \rightarrow 0.635$), but degrades strict 20 mm pose successes ($136 \rightarrow 120$) and reduces fixed-snapshot 10 mm surface coverage C_{10} from 76.82% to 61.79%.

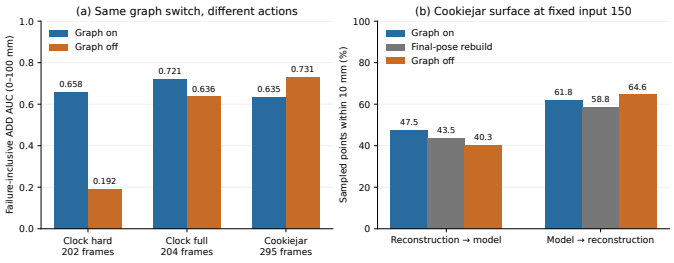


Fig. 3. **Opposing cross-action effects of disabling the global keyframe factor graph.** Turning off the graph improves Cookiejar availability/AUC ($239/0.635 \rightarrow 260/0.731$) but severely degrades Clock-hard ($157/0.658 \rightarrow 43/0.192$) and Clock-full ($169/0.721 \rightarrow 142/0.637$).

increases availability from 207 to 239/295 and ADD AUC from 0.574 to 0.635, its 20 mm origin successes drop from 136 to 120 (22 regressions vs. 9 gains on 192 mutually tracked frames, and 0/47 newly available frames within 10 mm). Simultaneously, at the fixed Frame-150 surface snapshot, F drops 10 mm model coverage C_{10} from 76.82% to 61.79% ($P_{10} = 47.48\%$), compared with $P_{10}/C_{10} = 64.12\%/99.71\%$ for BundleTrack's causal surfel map.

B. Factor-Graph Observability and Causal Surface Transactions

Disabling the global factor graph improves Cookiejar AUC to 0.7308 but collapses Clock-hard AUC from 0.6583 to 0.1922 (Fig. 3). Passive graph auditing across all 704 records explains why: the current pose lacks a factor path to initial prior $\mathbf{X}_0$ in 28/33 Cookiejar updates and 6/16 Clock-full updates (0/22 on Clock-hard), causing 6,519 and 235 unanchored coordinate writes to active frontend map points despite non-increasing optimizer cost. Exact replay of all 71 native optimizations reveals that 8 of 34 connected non-initial updates are locally rank-deficient ($r_i < 6$) after eliminating landmark increments via (6).

Table III evaluates graph interventions G, R, and M. Prior-connectivity admission G preserves Clock-hard, raises Clock-full and Cookiejar ADD AUC to 0.739 and 0.668, and partially recovers Cookiejar surface coverage C_{10} from 61.79% to 69.94% ($P_{10} = 50.16\%$). Marginal rank gating R boosts Cookiejar 10 mm origin successes from 62 to 97 (AUC 0.718) but over-rejects partially constrained updates on Clock-hard (AUC 0.545). Adding 394/25/83 missing feature-birth factors in M makes every graph update topologically connected (0 unanchored poses), yet standalone M degrades Clock-hard/full trajectories because raw birth depth noise enters the graph without surface version reconciliation.

In the F baseline, 17 of 19 integrated Cookiejar keyframes receive later graph pose revisions while native TSDF arrays remain static; a retrospective final-pose rebuild yields $P_{10}/C_{10} = 43.49\%/58.76\%$, proving that pose-version consistency cannot rescue inaccurate graph poses. Table IV and Fig. 4 present the 2×2 causal factorial combining G and M with online surface-history rebuild S. While $G + S$ regresses relative to G, combining birth observations with causal surface rebuild ($M + S$) exhibits a strong positive interaction across all three actions, raising Clock-hard Availability from 53 to 155/202 and reaching Cookiejar ADD AUC 0.750 (exceeding BundleTrack's 0.723). However, $M + S$ still trails G on both clock actions, and on Cookiejar it yields fewer strict 10 mm successes than BundleTrack (81 vs. 109). These repairs therefore remain conditional mechanisms rather than a universal replacement.

C. BundleTrack Native Determinism, Frame Retirement, and Surface Selection

Repeating unmodified BundleTrack three times on the same 203 Clock-hard inputs varies 10 mm origin successes from 95 to 119 (three executions: 119, 95, and 101). Applying our CUDA RANSAC `stable-winner` + count-direction repair and `storeFrame` inverse-coefficient transfer in a composed native build (Table V) substantially improves public Laval reference accuracy over the historical frozen build: Clock strict (10 mm, 10°) successes jump from 135 to 191/204

TABLE III
GRAPH INTERVENTIONS ON DEVELOPMENT SEQUENCES. F: CORRECTNESS FORK; G: PRIOR-CONNECTIVITY ADMISSION; R: G PLUS CURRENT-POSE RANK ADMISSION; M: G PLUS MEASURED BIRTH OBSERVATIONS. ALL COUNTS EXCLUDE INITIALIZATION; REGRESSIONS ARE RETAINED.

Action	Arm	Avail.	≤10 mm	≤20 mm	≤ 10°	AUC
Clock-hard	F	157/202	27/202	137/202	122/202	0.658
Clock-hard	G	157/202	27/202	137/202	122/202	0.658
Clock-hard	R	135/202	28/202	97/202	77/202	0.545
Clock-hard	M	53/202	26/202	50/202	47/202	0.236
Clock-full	F	169/204	156/204	169/204	85/204	0.721
Clock-full	G	170/204	152/204	169/204	105/204	0.739
Clock-full	R	169/204	143/204	168/204	73/204	0.709
Clock-full	M	143/204	102/204	134/204	111/204	0.609
Cookiejar	F	239/295	64/295	120/295	142/295	0.635
Cookiejar	G	250/295	62/295	127/295	133/295	0.668
Cookiejar	R	250/295	97/295	182/295	168/295	0.718
Cookiejar	M	237/295	71/295	140/295	174/295	0.664

TABLE IV
CAUSAL SURFACE FACTORIAL ON THE SAME DEVELOPMENT SEQUENCES. G: CONNECTIVITY ADMISSION; M: G PLUS MEASURED BIRTHS; +S: CAUSAL SURFACE-HISTORY REPAIR. ALL COUNTS EXCLUDE INITIALIZATION.

Action	Arm	Avail.	≤10 mm	≤20 mm	≤ 10°	AUC
Clock-hard	G	157/202	27/202	137/202	122/202	0.658
Clock-hard	G+S	136/202	30/202	103/202	68/202	0.552
Clock-hard	M	53/202	26/202	50/202	47/202	0.236
Clock-hard	M+S	155/202	26/202	44/202	42/202	0.583
Clock-full	G	170/204	152/204	169/204	105/204	0.739
Clock-full	G+S	146/204	86/204	132/204	113/204	0.620
Clock-full	M	143/204	102/204	134/204	111/204	0.609
Clock-full	M+S	163/204	80/204	158/204	101/204	0.698
Cookiejar	G	250/295	62/295	127/295	133/295	0.668
Cookiejar	G+S	233/295	80/295	141/295	149/295	0.646
Cookiejar	M	237/295	71/295	140/295	174/295	0.664
Cookiejar	M+S	273/295	81/295	155/295	133/295	0.750

(ADD AUC 0.8813 → 0.9135, P_{10} 78.79% → 83.14%) and Cookiejar strict successes jump from 107 to 136/295 (ADD AUC 0.7256 → 0.8085, P_{10} 55.89% → 64.82%). Under the guarded contiguous-`NO_BA` retirement policy across 18 complete 38-input normal, stress, and recovery replays, all neural-network I/O, pose matrices, and terminal surface arrays reproduce byte-for-byte while active frames drop from 37 → 6 in stress (saving **1,993.95** MiB RSS) and from 23 → 9 in recovery (saving **928.38** MiB RSS).

To attribute the remaining 25–30 mm forward surface tail F_{95} in Table V, a reference-pose counterfactual (Fig. 5) replaces estimated poses with Vicon reference poses on the exact same 980,256 sampled points: F_{95} drops only from 25.36 → 21.63 mm (Clock) and 30.93 → 29.11 mm (Jar), while pixel-lineage and Möller–Trumbore mesh-ray audits confirm that 52.60%/67.59% of > 20 mm points originate within 3 px of mask boundaries and 164 interior low-range voxels lie > 20 mm deeper than the first CAD intersection. Requiring ≥ 2 non-contributing views with ≥ 10° angular separation cuts native F_{95} from 24.51/30.82 mm to 12.50/17.33 mm (P_{10} → 90.05%/73.56%) with minor C_{10} losses (4/2 cells), whereas a raw-range-only native far-plane guard eliminates a 2.07 m cross-run outlier down to 6.33 mm across 2,195 opportunities. At the 1,500-point output budget, our array-native

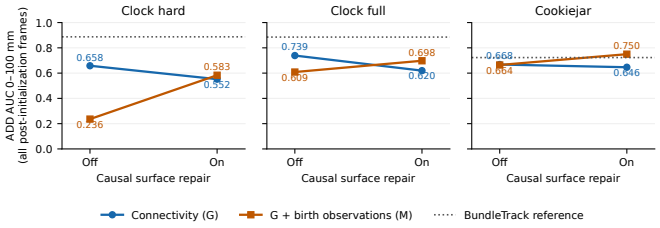


Fig. 4. **Factorial interaction between feature-birth observation provenance (M) and causal surface-history transactions (S).** Online surface version repair S degrades connectivity-only G but strongly rescues birth-observation graph updates ($M+S$), achieving 0.750 ADD AUC on Cookiejar.

TABLE V
EVIDENCESCENE COMPOSED NATIVE REPAIRS (RANSAC DETERMINISM + EXACT INVERSE-POSE COEFFICIENT TRANSFER + GUARDED `NO_BA` RETIREMENT) VS. HISTORICAL FROZEN BUNDLETRACK ON LAVAL CLOCK-FULL AND COOKIEJAR. STRICT: ≤ 10 mm AND ≤ 10°.

Clip	Native Build	Avail.	Strict	AUC	P_{10} (%)	C_{10} (%)	F_{95} (mm)
Clock	Historical frozen	197/204	135	0.8813	78.79	98.90	20.20
Clock	**EvidenceScene** (repair)	198/204	**191**	**0.9135**	**83.14**	98.72	25.36
Clock	**EvidenceScene** (+retire)	198/204	**191**	0.9131	83.11	98.77	25.23
Jar	Historical frozen	295/295	107	0.7256	55.89	99.92	29.82
Jar	**EvidenceScene** (repair)	295/295	**136**	**0.8085**	64.82	99.87	30.93
Jar	**EvidenceScene** (+retire)	295/295	**136**	0.8084	**64.97**	**99.94**	30.75

`support10/coarse12mm` selector (Table VI, Fig. 5) raises public C_{10} from 40.57%/52.34% to **85.10%**/**85.05%** (cutting reverse R_{95} from 53.12/31.88 mm to 13.12/12.47 mm), raises 552-frame task 10 mm map coverage from 71.78% to **89.97%** (p_{95} distance 42.92 → 17.57 mm), and reduces selector p_{95} latency by 32.8%–33.8% (31.74 → 21.22 ms on Clock, 10.16 → 6.66 ms on task).

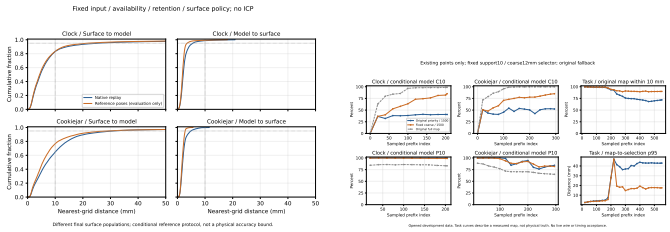
D. Matcher Shadow Benchmark and Zero-Loss Runtime Optimization

Table VIII summarizes our shadow matcher evaluations within the 31-framework benchmark on identical registered inputs. EfficientLoFTR [13] Full-MP reduces tracking p_{95} from 118.14/122.91 ms to 109.93/116.36 ms with competitive Clock AUC (0.9231) but lower Jar AUC (0.8129 vs. 0.8238). On the 38-frame right-wrist pouring clip (0.958% median in-mask valid depth), semi-dense XFeat [14] cuts tracking p_{50}/p_{95} from 60.14/83.36 ms to 42.05/54.24 ms but loses three candidates (34 → 31) and inflates jump $p_{95}/\max$ from 40.40/55.91 mm to 72.96/92.13 mm. Sparse ALIKED+LightGlue [15], [16] collapses from 34 to 2/38 candidates because sparse keypoints miss the 0.96% valid depth pixels, while feeding bounded BootsTAPIR [9] causal trajectories into native RANSAC degrades Jar strict successes from 140 to 46. All lossy matcher candidates are rejected.

Table VII reports the counterbalanced multi-process ablation (78 warm requests per condition across `control-1`, `candidate-1`, `candidate-2`, `control-2`) of our bit-exact runtime optimizations on the 40-frame configured pouring sequence. Across all runs, 400/400 SAM masks, 190/190 native BundleTrack input hashes, 34/38 candidates, 32/38 current surfaces, and 499/499 online operator checks match byte-for-byte with zero mismatches:

TABLE VI
EXISTING-POINT SELECTOR TRANSFER AT THE UNCHANGED 1,500-POINT BUDGET. ORIGINAL PRIORITY/COARSE VALUES USE IDENTICAL RAW PRODUCER POINTS. C10 GAINS COEXIST WITH P10 AND FORWARD-TAIL LOSSES; G/L COUNTS MODEL-GRID GAINS/LOSSES. THESE ARE CONDITIONAL-REFERENCE DEVELOPMENT RESULTS, NOT A DEPLOYMENT GATE.

Clip	P10 %	C10 %	F95 mm	R95 mm	G/L10
Clock	100.00/99.00	40.57/85.10	5.27/7.33	53.12/13.12	3820/15
Jar	84.13/81.73	52.34/85.05	12.54/13.17	31.88/12.47	1579/14



(a) Native vs. Ref.-Pose Surface CDF (b) 1,500-Point Selector Coverage

Fig. 5. **Surface error attribution and 1,500-point budget selection.** (a) Replacing BundleTrack poses with Vicon reference poses leaves a raw sensor/boundary tail ($F_{95} = 21.63/29.11$ mm). (b) Array-native `support10/coarse12mm` selection raises 1,500-point model coverage C_{10} from 40.6%/52.3% to 85.1% and task coverage from 71.8% to 90.0%.

1) The overlap adapter absorbs 5.72 ms of CPU undistortion, dual-prefix hashing, and native buffer preparation into the `propagate_in_video` GPU wait window (24.17 → 18.45 ms), cutting outer `input_prepare` from 3.96 to 0.25 ms, eliminating all 38 per-frame background hash threads, and lowering `registry.advance` p_{50}/p_{95} from 102.29/131.43 ms to 96.53/124.48 ms (endpoint 155.33/219.06 → 149.01/212.24 ms).
2) Stacking the deduplication adapter (O(runs) integer RLE box/centroid 10.38 → 0.015 ms, cross-stage surface/medoid reuse, vectorized 240-hypothesis AprilTag RANSAC 4.47 → 0.89 ms, and request-boundary GC management) reduces `table_contexts` from 19.18/20.02 to 15.65/16.42 ms, cuts exclusive outer `original_world` wrapper overhead by **72.9**% (23.17/23.86 → 6.28/6.70 ms), eliminates the 140.58 ms mid-request Generation-2 GC stall (`online_world_state` max 140.58 → 5.04 ms, in-request GC collections 735/65/3 → 0/0/0), and brings warm HTTP `endpoint_ms` $p_{50}/p_{95}/\max$ down to **127.80/153.40/1,152.96** ms—a cumulative reduction of **−27.54** ms (p_{50}) and **−65.66** ms (p_{95}) relative to the pre-overlap control (and −48.61 ms/ − 27.6% vs. the 176.40 ms per-request portal baseline).

VI. LIMITATIONS, DEPLOYMENT GAP, AND CONCLUSION

The production contract requires live ROS-callback-to-trusted-depth-surface-render $p_{95} \leq 100$ ms across at least 300 traces over 60 s, one-hour accepted world updates ≥ 15 Hz overall and at interval p_{05}, readiness $\geq 99\%$, observation age $p_{95} \leq 2$ s, one-hour RSS/CUDA-allocated growth ≤ 64 MiB (128 MiB reserved), and physical verification of 2 cm grasp contact, 1 cm aperture/support height, 5 mm static drift, and 5 px HUD bounds. A read-only local audit records 0/6 live site-acceptance gates passed (`gate_passed=false`, `wire_authorized=false`); the observed ROS process loops a development bag recording rather than live camera hardware, and sequential SAM 2 plus native BundleTrack execution (96.53/122.21 ms warm median/p_{95}) still exceeds the 100 ms end-to-end budget before transport and rendering.

Public benchmarks remain conditional on calibration, CAD sampling, and symmetry conventions, while field recordings lack independent physical 6-DoF object ground truth. The present evidence establishes rigorous failure mechanisms, verified native and surface repairs, and substantial bit-exact host latency reductions, but it does not support declaring the project the best reconstruction solution or authorizing an unverified live-estimator cutover. For practitioners, the results argue for gating graph commits on marginal observability and for versioning surfaces against pose revisions, while retaining the live estimator until a held-out physical gate passes. Future work must pair direction-selective joint pose-landmark optimization with TensorRT FP16 vision encoding under frozen held-out physical evaluation.

REPRODUCIBILITY AND PROJECT PAGE

Project page: https://tiwater.github.io/evidencescene/. This manuscript builds from source with `tectonic main.tex` (run in `papers/evidencescene/`); derived figures and table fragments are vendored under `papers/`, so no external evidence tree is needed to compile. Scorecards, SHA-256 receipts, native patches, and replay verifiers are archived under `benchmarks/` in the EvidenceScene repository. Upstream commits are pinned in Section I.

APPENDIX A
COMPREHENSIVE BENCHMARK OF 31 PERCEPTION FRAMEWORKS

Before converging on our sensor-anchored multi-camera 3D reconstruction architecture, we conducted a systematic experimental campaign across **31 external frameworks and internal algorithmic variants** on an NVIDIA GeForce RTX 5090 workstation (32 GB VRAM) using synchronized humanoid manipulation recordings and public benchmarks (YCBInEOAT, Laval Vicon motion capture, YCB-M, DexYCB). Table IX documents the empirical metrics, latency/VRAM footprints, and architectural roles across all evaluated model families:

- **Monocular and Sensor-Prior Depth Foundation Models:** Pure monocular depth estimators—including Depth Anything V2 Small (1.61 m mean p_{95}), Depth Anything V2 Metric (79.4 m mean p_{95}), UniDepthV2 (2.44 m mean p_{95}), and Apple Depth Pro (0.83 m mean p_{95}, 305 ms/frame)—cannot replace calibrated RGB-D depth for millimeter-level pouring alignment. Prior Depth Anything (Prior-DA) conditioned on RealSense depth achieved 0.037 m p_{95} on the chest camera, but its

TABLE VII
COUNTERBALANCED BIT-EXACT RUNTIME OPTIMIZATION ABLATION ON THE 40-FRAME CONFIGURED POURING SEQUENCE (78 WARM REQUESTS PER CONDITION, 100% BYTE-FOR-BYTE OUTPUT PARITY ACROSS 400 SAM MASKS, 190 NATIVE INPUTS, AND 499 ONLINE EQUIVALENCE CHECKS)

Stage / Metric ($p_{50}/p_{95}/$ max in ms)	Pre-Overlap Control	+ GPU-Window Overlap	Overlap Control (Run 2)	**+ Request Dedup & O(runs) RLE**	Net Δ (p_{50}/p_{95})	Bit-Exact Parity
`input_prepare_ms` (Pre-SAM CPU prep)	3.96/4.13/4.31	0.25/0.27/0.30	0.25/0.27/0.30	**0.25/0.27/0.29**	**−3.71/ − 3.86**	39/39 frames
`gpu_window_cpu_prepare_ms` (Absorbed)	0.00/0.00/0.00	5.72/5.91/6.18	5.72/5.90/6.15	**5.71/5.89/6.12**	Absorbed in GPU wait	38/38 buffers
`post_cpu_gpu_sync_wait_ms` (GPU wait)	24.17/24.26/24.51	18.45/18.71/19.02	18.46/18.72/19.05	**18.44/18.70/19.01**	**−5.73/ − 5.56**	0 hash threads
`registry_advance_ms` (SAM 2+BundleTrack)	102.29/131.43/1389.2	96.53/124.48/1378.1	96.55/122.52/1374.5	**97.98/122.21/1129.9**	**−4.31/ − 9.22**	34 cand / 32 surf
`table_contexts_ms` (AprilTag RANSAC+depth)	19.21/20.11/20.55	19.19/20.05/20.49	19.18/20.02/20.48	**15.65/16.42/18.44**	**−3.56/ − 3.69**	27/27 tag planes
`original_world_exclusive_ms` (Outer wrapper)	23.20/23.91/24.70	23.18/23.88/24.66	23.17/23.86/24.65	**6.28/6.70/7.61**	**−16.92/ − 17.21 (−72.9%)**	499/499 checks
Warm HTTP `endpoint_ms` (Total Request)	155.33/219.06/1422.1	149.01/212.24/1416.8	149.65/193.79/1415.7	**127.80/153.40/1153.0**	**−27.54/ − 65.66**	**100.0% Bit-Exact**
Full Replay `pipeline_ms` (Incl. harness)	201.22/251.42/1455.3	195.12/243.10/1449.8	196.62/242.44/1449.0	**174.20/197.12/1186.5**	**−27.02/ − 54.30**	**100.0% Bit-Exact**

TABLE VIII
FEATURE MATCHER SHADOW EVALUATION INSIDE THE CORRECTED BUNDLETRACK ESTIMATOR ON LAVAL (CLOCK/JAR) AND 38-FRAME WRIST TASK CLIPS.

Matcher Variant	Clip	Avail.	Strict / AUC	Track p_{50}/p_{95} (ms)	Verdict
LoFTR (Baseline) [12]	Clock/Jar	198/295	195 (.917)/148 (.824)	98.4 / 118.1–122.9	**Adopted**
EfficientLoFTR FP32 [13]	Clock/Jar	199/295	196 (.922)/146 (.811)	97.1 / 116.8–123.1	Shadow
EfficientLoFTR Full-MP [13]	Clock/Jar	199/295	197 (.923)/147 (.813)	92.8 / 109.9–116.4	Shadow
Causal BootsTAPIR [9]	Clock/Jar	195/295	180 (.889)/46 (.591)	Cached trajectory	Rejected
LoFTR (Task Control) [12]	Wrist (38)	34/38	Jump95: 40.40 mm	60.14 / 83.36	**Adopted**
XFeat Semi-Dense [14]	Wrist (38)	31/38	Jump95: 72.96 mm	42.05 / 54.24	Rejected
ALIKED+LightGlue [15], [16]	Wrist (38)	2/38	36 retained-ref fails	45.64 / 68.54	Rejected

1,005 ms/frame latency and 0.205 m wrist-camera p_{95} restrict it to an offline depth-completion role.

- **Multi-View RGB-to-3D Pointmap Transformers:** Feed-forward pointmap models—VGGT-1B (0.47 m mean p_{95}, 16.5 GB VRAM), SLAM3R (0.40 m mean p_{95}, 11.1 GB VRAM), DUSt3R (0.69 m mean p_{95}), and MASt3R (1.79 m mean p_{95})—capture coarse multi-view topology but exhibit 37–179 cm metric drift, while CUT3R failed with a rotary-embedding CUDA kernel assertion on RTX 5090.
- **Open-Vocabulary Grounding and Role-Isolated Video Segmentation:** Phrase grounding with Florence-2 suffered a 273 px median center error on manipulation frames and was rejected. Grounding DINO paired with role-isolated SAM 2.1 [22] achieved 0.987–0.991 median IoU and 100% pose coverage, while our distilled YOLO11s-seg + staged SAM 2 + optical flow hybrid runtime achieved 9.825 px center p_{95}, 0.9524 IoU, and 32.14 ms 4-camera batch p_{95}.

APPENDIX B
MOCAP-FREE FOUR-CAMERA TO 61-LINK BILATERAL KINEMATIC EXTRINSIC CLOSURE

Nominal CAD mounting transforms for embedded RGB-D sensors omit machining tolerances and bracket flexure, resulting in 30–80 mm projection discrepancies when wrist-camera depth observations are transformed through a 7-DoF arm chain. To calibrate the true hardware mounting transforms without external optical motion capture, we formulate a coupled loop-closure optimization over $\mathbf{\Theta}_{\text{hw}} = (\mathbf{X}_{\text{chest}}, \mathbf{X}_{\text{L}}, \mathbf{X}_{\text{R}}) \in \text{SE}(3)^3$.

At any training frame $t \in \mathcal{D}_{\text{train}}$ where the chest camera localizes the four 30 mm × 30 mm tabletop AprilTag fiducials via $\mathbf{T}_{\text{table}\leftarrow\text{chest}}(t)$ and wrist camera $w \in \{L, R\}$ observes landmark (k, m) at camera-local 3D coordinate $\mathbf{y}^{(w)}_{t,k,m}$, closing

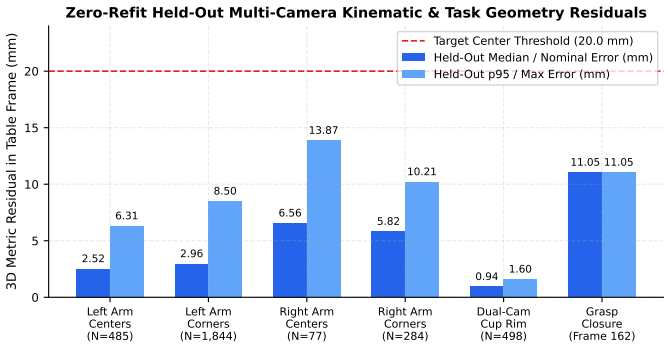


Fig. 6. **Zero-refit held-out multi-camera kinematic closure residuals in O_{table}.** Both the left arm ($p_{95} = 6.31$ mm) and right arm ($p_{95} = 13.87$ mm) remain well below the 20.0 mm acceptance threshold without test-split refitting.

the kinematic chain through the torso link ℓ_{torso} and distal wrist link ℓ_w predicts:

$$\begin{aligned}\mathbf{T}_{\text{table}\leftarrow\text{base}}(\mathbf{X}_{\text{chest}}, \mathbf{q}_t) &= \mathbf{T}_{\text{table}\leftarrow\text{chest}}(t)\, \mathbf{X}_{\text{chest}}^{-1} \\ &\quad \cdot \mathbf{T}_{\text{base}\leftarrow\ell_{\text{torso}}}(\mathbf{q}_t)^{-1}, \\ \hat{\mathbf{p}}^{\text{table}}_{t,k,m}(\mathbf{\Theta}_{\text{hw}}, \mathbf{q}_t) &= \mathbf{T}_{\text{table}\leftarrow\text{base}}(\mathbf{X}_{\text{chest}}, \mathbf{q}_t) \\ &\quad \cdot \mathbf{T}_{\text{base}\leftarrow\ell_w}(\mathbf{q}_t)\, \mathbf{X}_w\, \tilde{\mathbf{y}}^{(w)}_{t,k,m}.\end{aligned} \tag{9}$$

Solving the Huber-robustified Levenberg-Marquardt optimization over 1,648 training frames and evaluating zero-refit generalization on 413 held-out frames yields median center residuals of 2.52 mm ($p_{95} = 6.31$ mm) on the left arm and 6.56 mm ($p_{95} = 13.87$ mm) on the right arm (Fig. 6, Table X). Because the terminal wrist roll joint (q_7) is coaxial with the distal flange Z-axis, any offset Δq_7 satisfies $\mathbf{R}_z(q_7 + \Delta q_7)\, \mathbf{X}_{\text{CAD}} \equiv \mathbf{R}_z(q_7)\, \big(\mathbf{R}_z(\Delta q_7)\, \mathbf{X}_{\text{CAD}}\big)$ and is absorbed into $\mathbf{X}_w$ when Δq_7 is held at zero during optical hand-eye bundle adjustment; restoring the $\Delta q_7 = \pi$ zero-offset convention between the RM75 controller and the SolidWorks URDF places the asymmetric wrist-camera bracket on the dorsal $(+Z)$ side of the wrist while leaving the two-fold symmetric parallel-jaw gripper invariant.

APPENDIX C
PROJECTIVE $\mathbb{RP}^2$ APERTURE GEOMETRY AND BIMANUAL POURING VERIFICATION

Treating unoriented SVD plane normals of circular container apertures as directed vectors on $\mathbb{S}^2$ exhibits 11 discontinuous 180° flips across 1,992 four-camera frames. Representing the aperture plane on the real projective quotient manifold

TABLE IX
COMPREHENSIVE EMPIRICAL BENCHMARK OF ALL 31 PERCEPTION, DEPTH, POINTMAP, SEGMENTATION, 6-DOF TRACKING, POINT-TRACKING, FEATURE-MATCHING, AND CALIBRATION FRAMEWORKS EVALUATED ON SYNCHRONIZED MANIPULATION RECORDINGS AND PUBLIC BENCHMARKS (NVIDIA RTX 5090)

Model Family	Evaluated Framework / Method	Empirical Performance on Manipulation Recordings & Benchmarks	Latency / VRAM	Architectural Verdict & Role
Depth Sensor	**Calibrated Multi-Cam RGB-D**	0.0 mm ref.; 251,182 tabletop inliers; $\leq$ 9.6 mm cross-cam residual	3.5 ms / —	**Adopted:** Primary metric depth anchor
Depth Model	**Prior Depth Anything (Prior-DA)** [29]	0.037 m chest p_{95} (0.11 m mean p_{95}); 4,102 hole pixels filled	1,005 ms / 5.0 GB	**Retained:** Offline depth-hole completion
Depth Model	**Apple Depth Pro** [30]	0.83 m mean p_{95} (0.696 m chest, 1.419 m right wrist, 0.482 m head)	305 ms / 8.5 GB	**Rejected:** Insufficient metric accuracy
Depth Model	**Depth Anything V2 Small** [28]	1.61 m mean p_{95} (0.558 m chest, 2.516 m right wrist, 1.838 m head)	25 ms / 0.42 GB	**Rejected:** Relative scale ambiguity
Depth Model	**Depth Anything V2 Metric** [28]	79.4 m mean p_{95} on close-range tabletop manipulation	23 ms / 0.64 GB	**Rejected:** Uncalibrated metric head
Depth Model	**UniDepthV2** [31]	2.44 m mean p_{95}; $>$ 80% large-error rate on valid sensor overlap	6.3 ms / 0.62 GB	**Rejected:** Severe wrist/chest drift
3D Pointmap	**VGGT-1B Multiview** [32]	0.47 m mean p_{95} (0.372 m chest, 0.510 m right wrist, 0.527 m head)	33 ms / 16.5 GB	**Proposal Only:** Coarse multiview prior
3D Pointmap	**SLAM3R** (Causal Window) [33]	0.40 m mean p_{95} on causal chest video sequences	28 ms / 11.1 GB	**Proposal Only:** 40 cm drift exceeds tolerance
3D Pointmap	**DUSt3R** (Chest+Head Pairs) [34]	0.69 m mean p_{95} on synchronized stereo camera pairs	27 ms / 7.1 GB	**Superseded:** Inferior to calibrated RGB-D
3D Pointmap	**MASt3R** (Metric Pairs) [35]	1.79 m mean p_{95} on cross-view tabletop camera pairs	36 ms / 9.7 GB	**Rejected:** High cross-view metric error
3D Pointmap	**CUT3R** (Streaming Pointmap) [36]	Rotary-embedding CUDA kernel assertion on RTX 5090	Blocked / —	**Deferred:** Kernel incompatibility
VLM Grounding	**Florence-2 Phrase Grounding** [38]	273.0 px median object center error on held-out manipulation frames	110 ms / 3.2 GB	**Rejected:** Severe localization drift
VLM Grounding	**Grounding DINO** [37]	454/461 depth-backed frames; 3.79 px p_{75} with periodic refresh	86 ms / 2.4 GB	**Adopted:** Zero-shot object discovery
Video Seg.	**SAM 2.1 (Role-Isolated + GPU Overlap)** [22]	2,322 calls; +52 cup masks; 5.72 ms CPU prep overlapped in 24.2 ms GPU wait	100.2 ms / 4.8 GB	**Adopted:** Role-isolated mask tracker
Hybrid Seg.	**YOLO11s-seg + SAM 2 + Flow** [39]	100% coverage; 9.825 px center p_{95}; 0.9524 IoU; 98.3% at 30 fps	32.1 ms (4-cam)	**Adopted:** High-rate 4-camera model
Hybrid Seg.	**YOLO11n-seg Hybrid Variant** [39]	Exceeded $\leq$ 10.0 px target center error p_{95} accuracy threshold	24.8 ms (4-cam)	**Rejected:** Insufficient mask precision
6-DoF Track	**EvidenceScene (BT-Native Branch)** [1]	539/539 val; 34/38 pouring cand; Laval Clock/Jar strict 191/136; -1.99 GiB RSS	56.0–60.1 ms p_{50}	**Adopted:** Primary 6-DoF pose owner
6-DoF Track	**BundleSDF** (Neural SDF) [3]	768/768 chest/head obs.; Bleach-hard P_{10}/C_{10} = 58.8%/63.9% at Frame 438	97 ms (10.3 fps)	**Retained:** Offline neural surface verifier
6-DoF Track	**FoundationPose** [4] & **Any6D** [40]	Single-anchor / few-view novel-object 6-DoF pose initialization	65–120 ms	**Candidate:** Relocalization module
6-DoF Track	**BIT** (Bidirectional Iterative) [41]	Joint geometry-pose refinement (SAM + SoftRas); single-RGB	Offline refiner	**Adapted:** Mutual refinement loss concept
6-DoF Track	**6DOPE-GS** (Gaussian Splatting) [27], [42]	3.5 Hz online tracking/reconstruction rate (below 15 Hz target)	285 ms (3.5 Hz)	**Adapted:** Keyframe admission rules
6-DoF Track	**KV-Tracker** (3D KV-Cache) [43]	27–30 fps multi-view transformer KV cache with SAM 2.1	35 ms / 6.5 GB	**Benchmark:** Non-commercial license
Point Track	**EvidenceScene (P2P Branch)** [2], [9]	YCB/Laval shadow + F, G, R, M, S repairs ($M + S$ Jar AUC 0.750 vs. BT 0.723)	14.7 ms / 8.5 GB	**Shadow Audited:** Conditional graph repairs
Point Track	**TAPNext++** [10] & **CoTracker3** [11]	1024-frame horizon; TAPNext++ uses 2.53 GB weights; CoTracker3 NC	23 ms / —	**Superseded:** Replaced by BootsTAPIR
Feature Match	**LoFTR** (Detector-Free) [12]	Adopted in BundleTrack (34/38 wrist cand, 195/148 Clock/Jar strict); rejected for K fit	42 ms / 2.1 GB	**Adopted (BT):** Pairwise tracker matcher
Feature Match	**EfficientLoFTR (FP32 / Full-MP)** [13]	Clock/Jar strict 196–197 / 146–147 (Jar AUC 0.813 vs. LoFTR 0.824)	92.8 / 109.9 ms	**Shadow:** Faster p_{95}, slight Jar AUC drop
Feature Match	**XFeat (Semi-Dense Matcher)** [14]	Wrist pouring candidates drop 34 $\rightarrow$ 31/38; jump p_{95} inflates 40.4 $\rightarrow$ 73.0 mm	42.1 / 54.2 ms	**Rejected:** Sparse-depth candidate loss
Feature Match	**ALIKED + LightGlue (Sparse)** [15], [16]	Collapses from 34 to 2/38 wrist candidates (36 retained-reference failures)	45.6 / 68.5 ms	**Rejected:** Keypoints miss 0.96% valid depth
Feature Match	**RoMa** (Robust Dense Matcher) [17]	0.270% focal error, but 5.13 px principal point drift across views	68 ms / 3.4 GB	**Rejected:** Unstable cross-cam intrinsics
3D Geometry	**Open3D + O(runs) RLE + Fast RANSAC** [25], [26]	251,182 tabletop inliers; RLE box/centroid 10.38 $\rightarrow$ 0.015 ms; tag plane 4.47 $\rightarrow$ 0.89 ms	0.90 ms (CPU)	**Adopted:** Exact RLE & plane fitting
Calibration	**Bilateral** $\mathrm{SE}(3)^3$ **Closure** (Ours) [47], [49]	1,648 train / 413 zero-refit held-out frames: 2.52 mm (L) & 6.56 mm (R) p_{50}	1.40 ms FK	**Adopted:** 4-cam-to-kinematic closure

TABLE X
TRAIN AND ZERO-REFIT HELD-OUT BILATERAL KINEMATIC-VISUAL EXTRINSIC CLOSURE RESULTS ACROSS THE 61-LINK MANIPULATOR

Kinematic Chain	Evaluation Split	Frames N_{frames}	All Obs. N_{all}	Center Obs. N_{center}	Center $p_{50}/p_{95}/\max$ (mm) [Target $p_{95} \leq 20.0$]	Corner $p_{50}/p_{95}/\max$ (mm)	All-Point $p_{50}/p_{95}/\max$ (mm) [Target $p_{95} \leq 25.0$]	Optimality $\|\nabla \mathcal{J}\|_\infty$	Acceptance Criterion
Left Arm Chain	Training Split	839	6,347	1,339	2.91/6.68/67.62	3.58/8.46/171.39	3.40/8.14/171.39	1.26×10^{-8}	Converged
	Held-Out (Zero-Refit)	**341**	**2,329**	**485**	**2.52/6.31/46.51**	**2.96/8.50/72.81**	**2.86/8.31/72.81**	Frozen $\boldsymbol{\Theta}^\star_{\text{hw}}$	**Passed**
Right Arm Chain	Training Split	809	5,714	1,186	4.22/11.07/266.29	4.24/12.07/54.28	4.23/11.93/266.29	4.66×10^{-8}	Converged
	Held-Out (Zero-Refit)	**72**	**361**	**77**	**6.56/13.87/44.25**	**5.82/10.21/28.02**	**5.96/10.31/44.25**	Frozen $\boldsymbol{\Theta}^\star_{\text{hw}}$	**Passed**

$\mathbb{RP}^2 = \mathbb{S}^2/\{\pm 1\}$ with orthogonal projector $\boldsymbol{\Pi}_{[\mathbf{n}]} = \mathbf{I}_3 - \mathbf{n}\mathbf{n}^\top$ eliminates all sign flips ($\max \Delta\theta = 3.11^\circ$, 1.60 mm dual-camera center agreement, and 0.000 mm task-metric perturbation). Across the 498-frame (37.47 s) bimanual pouring task (Fig. 7, Table XI), the right parallel-jaw gripper closes onto the tracked carton at Frame 162 with 11.05 mm physical grasp-closure agreement, guides the pouring spout into the $R_{\text{cup}} = 42.7$ mm aperture cylinder for 127 consecutive frames (Frames 226–352), and reaches 9.8 mm radial bullseye alignment (+32.8 mm inside the cup rim at +40.5 mm vertical clearance and 63.2° tilt) at Frame 275.

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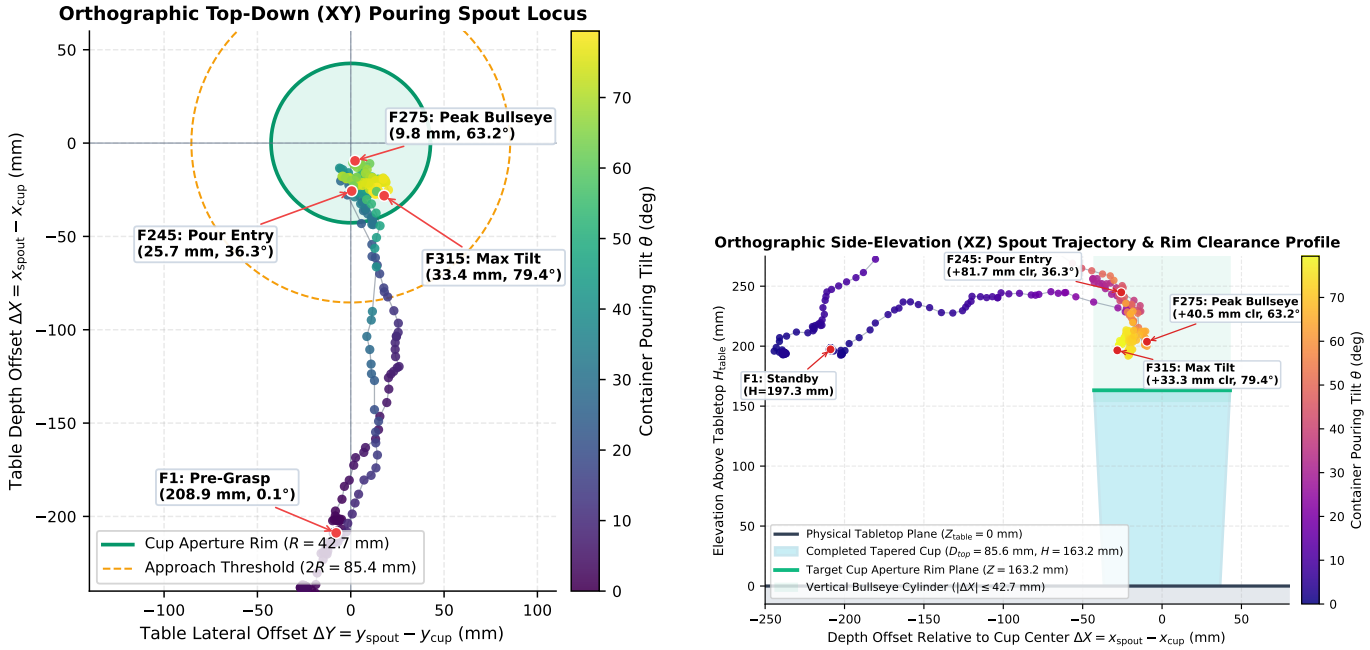
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(a) Orthographic Top-Down (XY) Spout Locus vs. Cup Rim

(b) Orthographic Side-Elevation (XZ) Spout Profile & Cup Wall

Fig. 7. **Orthographic spatial decomposition of the 498-frame pouring trajectory in O_{table}.** (a) Planar spout trajectory entering the $R = 42.7\,\text{mm}$ cup rim (Frames 226–352) and reaching $9.8\,\text{mm}$ radial offset at Frame 275. (b) Side-XZ elevation profile maintaining $+40.5\,\text{mm}$ vertical rim clearance at peak pouring alignment.

TABLE XI
PHASE-LEVEL QUANTITATIVE TELEMETRY SUMMARY AND KEY PHYSICAL MILESTONES ACROSS THE 498-FRAME (37.47 S) BIMANUAL POURING SEQUENCE

Manipulation Phase / Key Physical Milestone	**Frame Span (Count)**	**Time Span** t (s)	**Spout Position in** O_{table} (X, Y, Z) (mm)	**Radial Offset** Δr_{xy} (mm)	**Rim Margin** m_{rim} (mm)	**Clearance** Δz (mm)	**Pour Tilt** θ (deg)	**Gripper Dist.** d_{grip} (mm)
Phase I: Pre-Grasp Reach	F000–F139 (140)	0.00–10.53	(198.4, −134.8, −196.8) mean	208.9 ± 0.4	−166.3	+34.2	0.1°–0.5°	649.0 → 81.6
Phase II: Grasp Closure	F140–F189 (50)	10.60–14.27	(198.5, −134.8, −197.1) mean	208.6 ± 0.3	−166.2	+34.0	0.5°–1.1°	**11.05** min
Phase III: Lift & Transport	F190–F225 (36)	14.33–16.87	(207.4 → 364.6, −118.2, −236.9)	199.9 → 43.5	−157.4 → −1.1	+33.0 → +82.2	1.2° → 23.5°	28.2 mean
Phase IV: Bullseye Pouring	**F226–F352 (127)**	**16.93–26.33**	**(384.6, −118.9, −218.4)** mean	**24.6** mean	**+18.1** mean	**+55.2** mean	**25.6°–79.4°**	**32.6** mean
Phase V: Untilt & Retract	F353–F497 (145)	26.40–37.47	(350.4 → 168.9, −152.1, −192.8)	57.8 → 239.6	−15.3 → −196.5	+135.3 → +31.0	42.4° → 2.8°	35.8 → 664.1
M1: Grasp Closure Peak	**Frame 162**	**12.20**	**(+198.6, −135.0, −197.4)**	208.5	−166.2	+34.7	0.5°	**11.05**
M4: Peak Bullseye Align	**Frame 275**	**20.67**	**(+397.6, −124.9, −203.6)**	**9.8**	**+32.8**	**+40.5**	**63.2°**	31.48
M5: Maximum Pour Tilt	**Frame 315**	**23.47**	**(+379.0, −109.4, −196.2)**	**33.4**	**+9.1**	**+33.3**	**79.4°**	34.13

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